Kinect-like depth denoising: median filtering vs. Fu et al. (2012), evaluated by camera tracking on ICL-NUIM and TUM RGB-D

Kinect Lite project

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Abstract

We compare our per-frame depth cleaner (0.4 m gate + 9×9 median, optionally + inpainting) with the spatial-temporal pipeline of Fu et al. (hole-value prediction + ST-DNBL), in three readings of its hole step. Each cleaned sequence is tracked with depth-only point-to-plane ICP and with RGB-D odometry, and the paths are scored against ground truth. **Results:** on ICL the median halves the depth MAE (18.8 vs 39.2 mm input; paper 37.7 mm). For depth-only ICP the paper pipeline cuts drift 4–16× (ATE 0.95 vs 14.94 cm ICL, 3.31 vs 13.32 cm TUM). For RGB-D odometry the median is best (0.98 vs 26.0 cm ICL) or level (2.08–2.14 cm TUM). Of eight hole fills tested after the median, none improves RGB-D odometry on either dataset; the Telea inpainting we used is the least accurate of them (87 mm hole MAE on ICL vs 12–23 mm for the others). The three paper readings are indistinguishable.

1 Setup

Data	**ICL-NUIM** living room kt0, frames 1–300 (tracked 1, 3, …, 299): synthetic, noisy depth as input; noise-free depth and true trajectory as ground truth. **TUM RGB-D** fr3 long_office_household, frames 0–299 (tracked 0, 2, …, 298): real Asus Xtion; motion-capture trajectory, matched to each depth frame by nearest timestamp; no true depth.
Methods	**gate only**: depth < 0.4 m set to hole (baseline). 9×9 **median** (ours, `clean.py`): median over measured pixels, holes kept. **+ fill**: then OpenCV Telea inpainting, clamped to the local depth range. **paper** (`paper_pipeline.py`): spatial padding, block-matched hole prediction from frames $t \pm 2$ (Eq. 7–9), ST-DNBL over $t \pm 1$ with range kernel scaled by $\Theta(z) = (0.002z^2)^2$. The $i = 0$ term of Eq. 7 is read three ways: `paper` (as written), `exclude` (skipped), `nonlocal` (searched at non-zero shifts only).
Paper settings	$m = 2$, $r = 1$, search radius 6 px, $k = 9$, $\sigma_r = 16$, $\sigma_{\mathrm{MSE}} = 10^{-4}\,\mathrm{m}^2$, voids ≥ 100 px left empty; tuned on ICL frames 600–1400 (disjoint from the test frames).
Tracking	Every 2nd frame (150 frames, $\Delta t = 1/15$ s), frame to frame, motions chained into a path. **Depth-only ICP**: Open3D point-to-plane, three levels (voxel 4/2/1 cm, max correspondence 10/5/2 cm, 30 iterations each), initialised with the previous motion. **RGB-D odometry**: Open3D hybrid (photometric + geometric) term, identity initialisation.
Compute	ICL locally; TUM frames restored on Kaggle (4 cores), tracked locally. Restoring 450 frames: < 1 min for the three median-based methods, about 26 min for the paper pipeline.

2 Metrics

Metric	Definition	Reading
MAE (mm)	$\frac{1}{n}\sum \lvert d - d^* \rvert$ over pixels valid in both	typical per-pixel depth error
RMSE (mm)	$\sqrt{\frac{1}{n}\sum (d - d^*)^2}$	weights large misses; RMSE $\gg$ MAE signals spikes
median (mm), <1 cm (%)	median of $\lvert d - d^* \rvert$; share of pixels with $\lvert d - d^* \rvert < 10$ mm	robust to spikes; higher is better for <1 cm
ATE (cm)	$\sqrt{\frac{1}{N}\sum_k \lVert R\,p_k + t - p_k^* \rVert^2}$, with (R, t) the best rigid fit (Kabsch, no scale)	global drift of the whole path
RPE (mm, deg)	RMS over steps of $E_k = (T_k^{*-1} T_{k+1}^*)^{-1} (T_k^{-1} T_{k+1})$: $\lVert t(E_k) \rVert$ and $\angle R(E_k)$	local accuracy of one step; does not accumulate

Depth metrics are computed per frame and averaged over 300 ICL frames (d cleaned, d^* true depth); holes a method leaves empty are not scored. For scale: the true camera travels 0.97 m on ICL (6.5 mm per step) and

3.09 m on TUM (20.7 mm per step), so an RPE of 11.8 mm on ICL exceeds the motion being estimated. Differences below about 0.05 cm ATE are noise: a rerun on another machine moved TUM values by 0.01 cm.

3 Results

Method	MAE (mm)	RMSE (mm)	median error (mm)	within 1 cm (%)
gate only (input)	39.19	187.45	13.33	42.19
median 9×9 (ours)	**18.77**	**55.41**	**12.32**	**44.66**
median 9×9 + fill (ours)	19.34	57.67	12.36	44.47
paper (`center=paper`)	37.69	186.00	13.42	33.69
paper (`center=exclude`)	37.69	186.00	13.41	33.69
paper (`center=nonlocal`)	37.69	186.01	13.41	33.69

Table 1: ICL depth error against the noise-free depth, mean over 300 frames. Bold: best per column.

	ICL-NUIM kt0				TUM fr3 long_office_household			
	depth-only ICP		RGB-D odometry		depth-only ICP		RGB-D odometry	
Method	ATE	RPE	ATE	RPE	ATE	RPE	ATE	RPE
true depth (reference)	0.19	0.07	0.81	2.18	–	–	–	–
gate only (input)	17.07	14.43	23.59	24.20	13.97	11.69	2.14	**5.61**
median 9×9 (ours)	14.94	11.78	**0.98**	**1.59**	13.32	10.86	**2.08**	5.69
median 9×9 + fill (ours)	15.18	12.01	1.44	1.91	20.20	12.46	3.13	5.69
paper (`center=paper`)	**0.95**	**2.02**	26.07	21.76	3.33	**4.37**	2.10	5.80
paper (`center=exclude`)	**0.95**	**2.02**	26.03	21.76	**3.31**	**4.37**	2.10	5.79
paper (`center=nonlocal`)	**0.95**	**2.02**	26.01	21.75	**3.31**	**4.37**	2.12	5.80

Table 2: Camera path error, 150 frames: ATE in cm, translational RPE in mm (rotational RPE in `denoise/results/*_compare.txt`). The true-depth row tracks the noise-free ICL depth: the floor set by each tracker.

Depth. The ICL noise is dominated by spikes. The median removes them (RMSE $187 \to 55$ mm) and improves all 300 frames. The DNBL weights a neighbour by its similarity to the centre pixel, so a spike averages only with itself and survives (RMSE 186 mm). Its error is smooth but grows with distance, as the range kernel widens with $\Theta(z)$; fewer pixels end within 1 cm (33.7 vs 44.7 % for the median).

Depth-only ICP. The paper pipeline wins on both datasets (ATE 0.95 vs 14.94 cm ICL, 3.31 vs 13.32 cm TUM; RPE 2.0 vs 11.8 mm ICL). Point-to-plane ICP discards distant correspondences, so surviving spikes cost little, while it depends on surface normals. The DNBL yields smooth surfaces; the median leaves the striped, structured error of the sensor model, which biases the alignment systematically (no step failed; the error is spread over all steps). On ICL a spatial-only DNBL ($m = r = 0$) tracks even better (ATE 0.47 cm): the gain is the smoothing, not the temporal steps.

RGB-D odometry. The median is best on ICL (0.98 cm, close to the 0.81 cm floor); the paper pipeline fails (26.0 cm, worse than the input), because the odometry warps every pixel by its depth and uses depth gradients, so surviving spikes corrupt it (49 of 149 steps above 20 mm vs none for the median). On TUM all methods except + fill land within 2.08–2.14 cm. This fits real sensor noise being less spike-dominated than ICL's simulated noise, but we did not test that directly.

Hole filling and the hole-step readings. Inpainted values are invented geometry: + fill (Telea) is worse than the plain median in every tracking column (TUM depth-only 20.20 vs 13.32 cm); Section 4 compares seven other fills. The three readings of Eq. 7 differ by ≤ 0.06 cm: holes are about 1 % of pixels, so the choice is immaterial for tracking (see `icl_report.pdf` for the per-hole comparison).

4 Hole-filling methods

Eight fills applied after gate + 9×9 median (`experiments/hole_filling/compare_fills.py`, run on Kaggle): **none**; **Telea** and **Navier–Stokes** (OpenCV inpainting, clamped to the local depth range); **nearest** measured

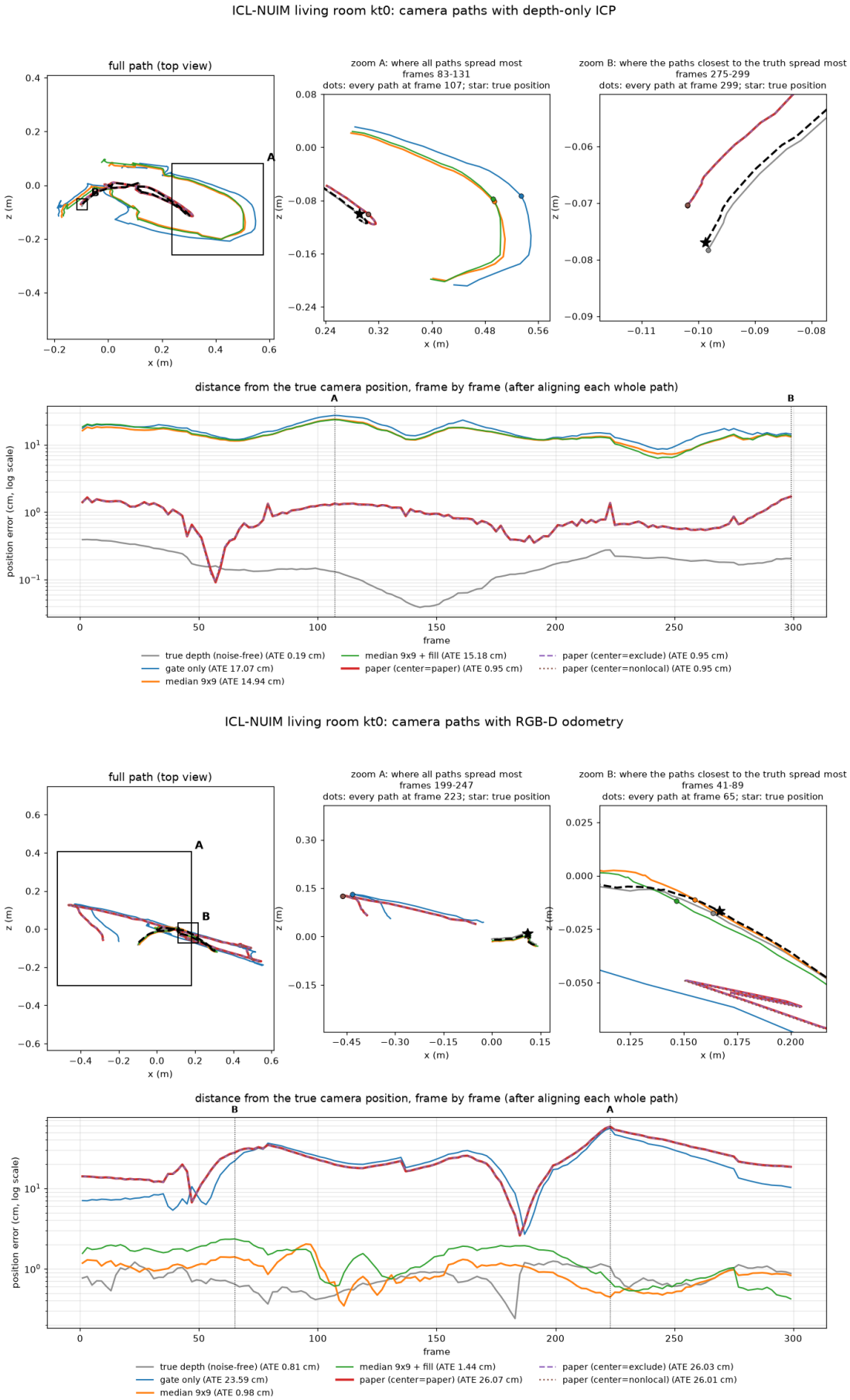
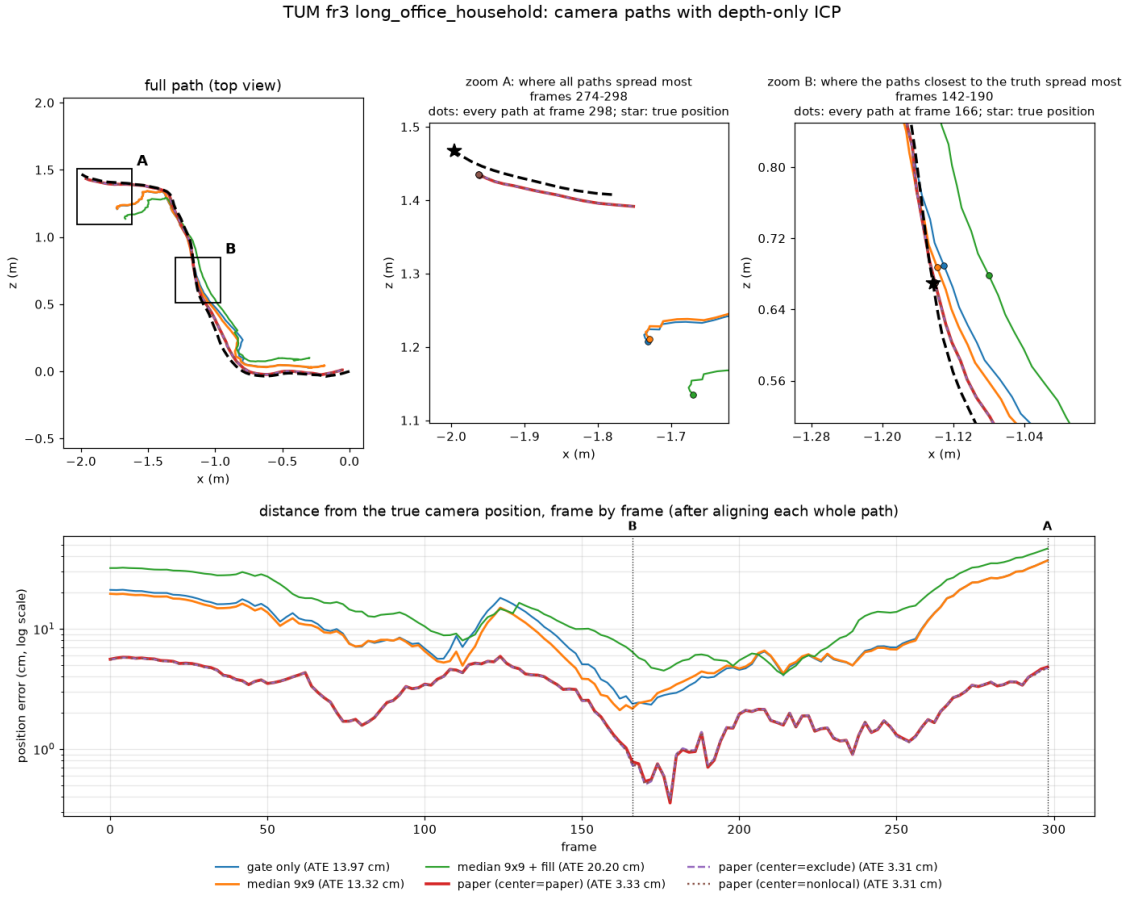


Figure 1: ICL camera paths: depth-only ICP (top), RGB-D odometry (bottom). Each panel set: full top view; zoom A where all paths spread most; zoom B where the paths near the truth (ATE $\leq 2\times$ best, plus the reference) spread most, drawn for the frame range in its title (dots: positions at one frame, star: truth); bottom: position error per frame, log scale. The three paper readings coincide (solid, dashed, dotted).



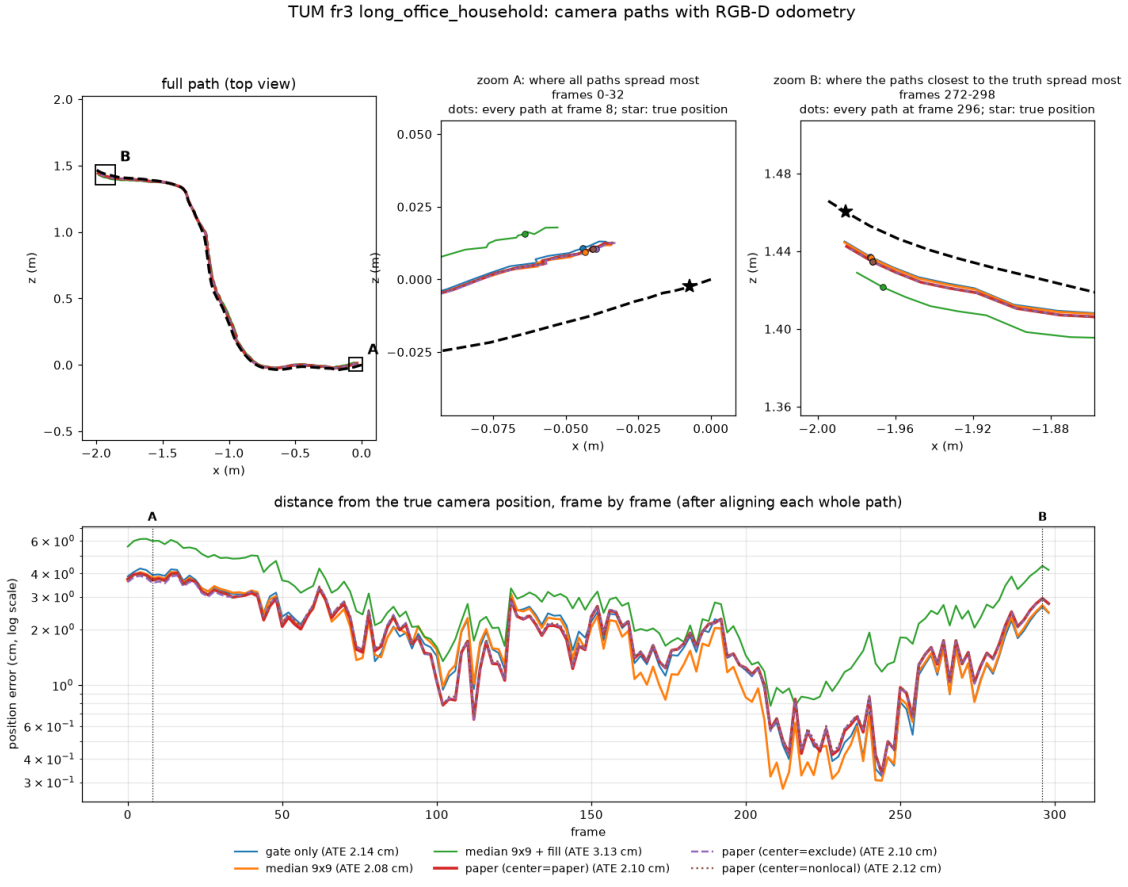


Figure 2: TUM camera paths: depth-only ICP (top), RGB-D odometry (bottom); panels as in the previous figure.

pixel; **region, earlier** (the earlier study's fill: small interior holes only, skipped at depth edges); **region median** (holes grow inward, each pixel the median of ≥ 3 known pixels in its 5×5 window); **background** (largest depth on the hole border, since Kinect holes are mostly the shadow behind a foreground edge); **temporal** (the paper's block-matched prediction from $t \pm 2$, no spatial term). Fill accuracy is scored two ways: on ICL against true depth at the real holes (2574 px, 0.84 % per frame), and on both datasets by a *masked test*: holes punched into measured pixels (992 px ICL, 10725 px TUM per frame) must be restored to the measured value. TUM has 16 % holes after the median, ICL under 1 %.

	ICL, real holes vs. true depth			ICL, masked test		TUM, masked test			
Fill after the median	MAE (mm)	median (mm)	<5 cm (%)	MAE (mm)	<5 cm (%)	MAE (mm)	median (mm)	<5 cm (%)	filled (%)
none (holes kept)	–	–	–	–	–	–	–	–	–
Telea (FMM)	87.0	36.8	62.2	234.5	45.2	626	501.6	14.7	**100.0**
Navier–Stokes	22.5	11.9	96.8	6.8	99.0	121	10.5	82.6	**100.0**
nearest	22.3	11.2	97.1	4.9	99.3	112	4.6	86.4	**100.0**
region, earlier (small holes)	15.5	**7.3**	95.8	**1.3**	**99.9**	11	1.6	97.7	23.2
region median (all holes)	22.5	12.6	97.3	5.8	99.3	117	7.2	85.4	**100.0**
background (far side)	83.1	43.1	58.9	266.9	43.0	3239	3529.1	11.6	**100.0**
temporal, paper (no spatial term)	**12.2**	8.2	**98.8**	4.4	99.7	**9**	**1.3**	**98.1**	39.0

Table 3: Fill accuracy, mean over frames. *filled*: share of masked TUM pixels the method fills (Telea, Navier–Stokes, nearest, region median and background always fill 100 %; on ICL's real holes region, earlier fills 4 % and temporal 88.5 %). Bold: best per column.

	ICL-NUIM kt0				TUM fr3 long_office_household				
	depth-only ICP		RGB-D odom.		depth-only ICP		RGB-D odom.		time
Fill after the median	ATE	RPE	ATE	RPE	ATE	RPE	ATE	RPE	(ms)
none (holes kept)	**14.94**	11.78	**0.98**	1.59	13.33	10.86	**2.08**	5.69	**0**
Telea (FMM)	15.18	12.01	1.44	1.91	20.21	12.46	3.13	5.69	210
Navier–Stokes	15.36	11.81	1.46	1.98	**12.03**	**10.71**	3.89	6.06	224
nearest	15.05	**11.72**	1.43	1.97	17.54	12.73	4.16	5.98	50
region, earlier (small holes)	**14.94**	11.77	1.01	**1.56**	13.32	10.87	2.23	5.67	52
region median (all holes)	15.15	11.76	1.43	1.99	15.14	11.87	4.08	5.98	386
background (far side)	15.24	12.05	1.80	2.10	13.51	10.94	3.14	**5.33**	40
temporal, paper (no spatial term)	15.11	11.88	1.24	1.62	13.37	10.92	2.51	5.57	7884

Table 4: Camera path error with each fill, 150 frames: ATE in cm, translational RPE in mm; time per frame for the fill alone.

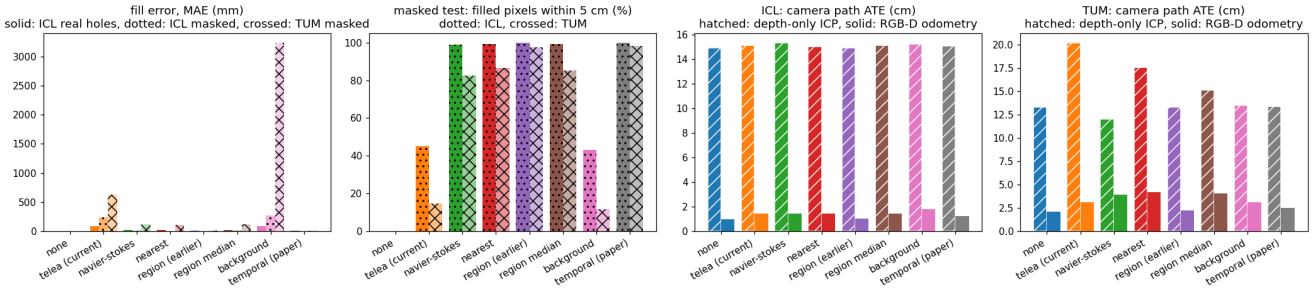


Figure 3: Fill error (left two) and camera path ATE (right two) for each fill.

Accuracy. Telea and background are far worse than the rest (ICL hole MAE 87 and 83 mm, TUM masked MAE 626 and 3239 mm): Telea blends both sides of a depth edge, background takes the far side even where the hole lies on the near surface. Navier–Stokes, nearest and region median fill every hole with about 22 mm MAE on ICL and 97 % within 5 cm. The temporal fill is the most accurate on ICL's real holes (12.2 mm, 98.8 % within 5 cm) but leaves 11.5 % unfilled and costs about 7.8 s per frame; region, earlier is accurate only because it fills just the easy, small holes.

Tracking. No fill improves RGB-D odometry: ATE without filling is 0.98 cm (ICL) and 2.08 cm (TUM), the best fill 1.01 and 2.23 cm, both from the fills that fill least. For depth-only ICP only Navier–Stokes on TUM helps (12.03 vs 13.33 cm), and the same fill is worse on ICL (15.36 vs 14.94 cm); one sequence each is not enough to call that a gain. A more accurate fill does not give better tracking: the trackers do better ignoring holes than using estimated depth.

Conclusion. For an RGB-D system use the 9×9 median without hole filling: best depth accuracy, best or equal-best tracking, negligible cost. The paper pipeline pays off only for depth-only tracking, where its spatial smoothing reduces drift 4–16×; on ICL, a cheaper spatial-only DNBL did even better. Where filled depth is needed (e.g. for display), replace Telea with Navier–Stokes or nearest: about 4× lower hole error on ICL.

Reproduce: `denoise/icl_compare.py`, `denoise/tum_compare.py` (frames 0–299, `--step 2`); hole fills: `experiments/hole_filling/compare_fills.py --frames 300 --step 2`; figures: `denoise/plot_paths.py`; report: `python report/build.py`, which regenerates the tables from `denoise/results/*_compare.txt` and `experiments/hole_filling/results/fills.txt`.